THREATS TO INFORMATION SECURITY

A Research-Based Survey and Analysis

Covering: Social Engineering • Malware • Network Attacks • Application Attacks • Infrastructure Threats

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Execution Summary

This survey presents a comprehensive research-based analysis of threats to information security, drawing on data from industry reports including Verizon DBIR, IBM X-Force, Ponemon Institute, Gartner, and ENISA Threat Landscape reports (2015–2024). The study identifies 25 distinct threats across five domains — social engineering, malware, network attacks, application attacks, and operational/infrastructure threats — evaluates each using a structured risk scoring model, and examines global spending trends and attack pattern evolution over the past decade.

Key findings indicate that Advanced Persistent Threats (APTs), Supply Chain Attacks, and Zero-Day Exploits represent the highest overall risk due to their high uncertainty, high asset value targeting, and relative insufficiency of current control mitigation. Phishing and ransomware, while more familiar, remain among the most operationally damaging threats due to their frequency. Cloud-based attacks and supply chain compromises are the fastest-growing threat categories between 2015 and 2024. Global cybersecurity spending is projected to exceed $300 billion by 2026, with no single threat category receiving insufficient investment relative to its risk score.

1. Identification of Threats

1.1 Overview

Twenty-five distinct information security threats were identified across five major domains. Each threat was assessed for its attack vector, behavioral characteristics, and notable real-world incidents. The table below provides a structured reference for all 25 threats, including brief descriptions and documented real-world examples drawn from widely reported incidents.

The five threat domains examined are: (1) Social Engineering — attacks exploiting human psychology rather than technical vulnerabilities; (2) Malware — malicious software designed to disrupt, damage, or gain unauthorized access to systems; (3) Network Attacks — threats targeting network infrastructure, communications, and protocols; (4) Application Attacks — exploits targeting software applications, databases, and APIs; and (5) Operational/Infrastructure Threats — threats arising from organizational processes, supply chains, physical environments, and cloud configurations.

Table 1.1 — 25 Identified Information Security Threats

#	Threat Name	Category	Description	Real-World Example	Primary Asset
1	**Phishing**	Social Engineering	Deceptive emails trick users into revealing credentials or clicking malicious links	2016 DNC spear-phishing: John Podesta's Gmail compromised	User Credentials
2	**Spear Phishing**	Social Engineering	Targeted phishing using personal info for higher success	2020 Twitter Bitcoin scam via employee phone spear-phishing	User Credentials
3	**Vishing**	Social Engineering	Voice-based social engineering over phone to extract sensitive data	2022 Uber breach: Attacker impersonated IT support to steal VPN creds	User Credentials
4	**Ransomwar e**	Malware	Encrypts victim files and demands ransom payment for decryption	2021 Colonial Pipeline: DarkSide ransomware, $4.4M ransom paid	Data / Operations
5	**Spyware**	Malware	Silently monitors user activity and exfiltrates sensitive data	Pegasus spyware (NSO Group) targeting journalists globally (2021)	Data / IP
6	**Rootkit**	Malware	Provides persistent privileged access while hiding from the OS	Sony BMG rootkit (2005): Silently installed on millions of PCs	Servers / OS
7	**Trojans**	Malware	Malicious code disguised as legitimate software	Emotet banking Trojan infected hundreds of thousands of systems 2014–2021	Data / Financial
8	**Worms**	Malware	Self-replicating malware spreading across networks autonomously	WannaCry (2017): 200,000+ systems in 150 countries via EternalBlue	Network Infrastructure
9	**Keylogger**	Malware	Records keystrokes to steal passwords and sensitive information	Dark Hotel APT used keyloggers on hotel Wi-Fi to steal executive credentials	User Credentials / Data
10	**SQL Injection**	Application Attack	Inserts malicious SQL code into input fields to manipulate databases	2008 Heartland breach: SQL injection exposed 130M card records	Data / Databases
11	**XSS**	Application Attack	Injects malicious scripts into trusted websites viewed by other users	2018 British Airways Magecart attack: 500,000 payment cards skimmed	User Credentials / Data
12	**Zero-Day Exploits**	Application Attack	Attacks exploiting unknown vulnerabilities before patches are available	Stuxnet (2010): 4 zero-days used to destroy Iranian nuclear centrifuges	Critical Infrastructure
13	**Man-in-the-Middle**	Network Attack	Secretly intercepts and relays communications between two parties	2011 DigiNotar CA compromise: Fraudulent SSL certs for 300,000 Gmail users	Network / Data
14	**DDoS**	Network Attack	Overwhelms target servers with traffic making services unavailable	2016 Mirai botnet attacked Dyn DNS, taking down Twitter & Netflix	Network Infrastructure
15	**DNS Spoofing**	Network Attack	Corrupts DNS cache to redirect users to malicious websites	2019 Sea Turtle campaign: DNS hijacking targeting government agencies	Network Infrastructure
16	**APT**	Network Attack	Long-term stealthy intrusion by sophisticated actors to exfiltrate data	SolarWinds (2020): Russian SVR compromised 18,000+ organizations	IP / Data
17	**Insider Threat**	Operational	Malicious or negligent actions by authorized users	2013 Snowden: NSA contractor exfiltrated	IP / Data

#	Threat Name	Category	Description	Real-World Example	Primary Asset
				classified surveillance documents	
18	**Supply Chain Attack**	Operational	Compromises vendor software/hardware to implant malicious code	SolarWinds Orion backdoor distributed to 18,000 customers (2020)	Software / Infrastructure
19	**Credential Stuffing**	Network Attack	Automated use of breached credentials to gain unauthorized access	2020 Zoom: 500,000+ accounts compromised via credential stuffing	User Credentials
20	**Cloud Misconfiguration**	Infrastructure	Improper cloud settings expose data or services to the public internet	2019 Capital One: Misconfigured AWS WAF exposed 100M records	Cloud Services / Data
21	**Social Media Exploitation**	Social Engineering	Using social platforms for OSINT gathering and social engineering	SolarWinds recon: Attackers used LinkedIn to target key IT personnel	User Credentials / Data
22	**Password Attacks**	Application Attack	Systematically guessing passwords via brute force or dictionary attacks	2012 LinkedIn breach: 6.5M MD5-hashed passwords cracked	User Credentials
23	**Physical Security Breach**	Operational	Unauthorized physical access to install rogue hardware or steal devices	2008 Hannaford supermarkets: Skimmers on POS terminals, 4.2M cards stolen	Physical Assets / Data
24	**IoT Exploitation**	Infrastructure	Attacks targeting insecure IoT devices to gain network access	2016 Mirai: 600,000 IoT devices hijacked to launch 1 Tbps DDoS	Network Infrastructure
25	**API Vulnerabilities**	Application Attack	Exploiting poorly secured APIs to access or manipulate backend data	2021 Experian: SSNs exposed via unauthenticated API endpoint	Data / Applications

Sources: MITRE ATT&CK Framework, OWASP Top 10 (2021), ENISA Threat Landscape 2023, Verizon DBIR 2023, IBM X-Force Threat Intelligence Index 2023

2. Threat Risk Evaluation and Ranking

2.1 Risk Scoring Methodology

Each threat was evaluated using the following risk model:

Risk Formula

Risk Score = (Likelihood × Asset Value × (1 − Control Mitigation%)) + UncertaintyWhere: Likelihood (L) = 1–10 | Asset Value (V) = 1–10 | Control Mitigation (CM) = 0–100% | Uncertainty (U) = 1–10

This model captures the fundamental security principle that risk is proportional to the probability and impact of an event, reduced by the effectiveness of deployed controls, with an additional term for uncertainty in vulnerability knowledge. Control mitigation values reflect the general effectiveness of industry-standard defenses rather than any single organization's controls.

2.2 Risk Score Results and Rankings

The following table presents all 25 threats ranked from highest to lowest risk score. Risk levels are color-coded: Critical (≥50), High (35–49), Medium (20–34), and Low (<20).

Table 2.1 — Threat Risk Evaluation and Ranking

Threat	Target Asset	Likelihood	Asset Value	Control Mit. %	Uncertainty	Risk Score	Level
Supply Chain Attack	Software/Infrastructure	7	10	30	9	**57.9**	CRITICAL
Zero-Day Exploits	Critical Infrastructure	7	9	20	9	**59.4**	CRITICAL
Advanced Persistent Threat	IP / Data	8	10	35	9	**61**	CRITICAL
Ransomware	Data / Operations	9	9	40	8	**56.6**	CRITICAL
Phishing	User Credentials	9	8	55	6	**38.2**	HIGH
Cloud Misconfiguration	Cloud Services / Data	8	9	45	7	**42.6**	HIGH
Insider Threat	IP / Data	7	9	50	8	**39.5**	HIGH
Spear Phishing	User Credentials	8	8	50	6	**38**	HIGH
IoT Device Exploitation	Network Infrastructure	7	7	45	7	**33.95**	MEDIUM
SQL Injection	Data / Databases	8	8	60	5	**30.6**	MEDIUM
Worms	Network Infrastructure	7	8	55	6	**31.2**	MEDIUM
API Vulnerabilities	Data / Applications	7	8	50	6	**34**	MEDIUM
Credential Stuffing	User Credentials	8	7	55	6	**31.2**	MEDIUM
Trojans	Data / Financial	7	8	60	5	**27.6**	MEDIUM
DDoS	Network Infrastructure	8	7	65	6	**25.6**	MEDIUM
Spyware	Data / IP	7	8	60	5	**27.6**	MEDIUM
MitM	Network/Data	6	8	60	6	**28.8**	MEDIUM
XSS	User Credentials/Data	7	7	60	5	**24.6**	MEDIUM
DNS Spoofing	Network Infrastructure	6	7	55	6	**24.9**	MEDIUM
Password Attacks	User Credentials	8	7	70	4	**20.8**	MEDIUM
Keylogger	User Credentials/Data	6	7	65	5	**19.7**	LOW
Rootkit	Servers / OS	5	8	60	6	**22**	MEDIUM
Social Media Exploitation	Credentials/Data	6	6	55	5	**21.2**	MEDIUM
Vishing	User Credentials	6	7	65	5	**19.7**	LOW
Physical Security Breach	Physical Assets/Data	5	7	60	5	**19**	LOW

CM% = approximate effectiveness of commonly-used cybersecurity defenses against each threat. Source: NIST SP 800-30, ENISA Threat Landscape 2023, Gartner Security Reports 2023.

2.3 Key Risk Findings

The four Critical-rated threats — APT, Zero-Day Exploits, Supply Chain Attacks, and Ransomware — share a common profile: high asset value targeting, relatively low control mitigation effectiveness, and high uncertainty. APTs achieve the highest scores because they target the most sensitive assets (intellectual property, classified data) while operating with stealth over extended periods, making them inherently difficult to detect and mitigate. Control mitigation for APTs is rated at only 35% because even organizations with mature security programs have repeatedly fallen victim (e.g., SolarWinds, Microsoft Exchange). Supply Chain Attacks score critically high because the control mitigation rate is extremely low (30%) — vendors and software update channels are implicitly trusted, leaving a wide attack surface. Zero-Day Exploits score highest on uncertainty (9/10) because, by definition, no patch or known defense exists at the time of exploitation. The high uncertainty term in the risk formula dramatically elevates their score despite their lower frequency compared to phishing.

3. Estimated Organizational Spending

3.1 Global Cybersecurity Investment Overview

Global cybersecurity spending reached approximately $188.3 billion in 2023 (Gartner) and is forecast to exceed $300 billion annually by 2026. Spending is driven primarily by the rising frequency and sophistication of the threats identified in this study, regulatory compliance requirements (GDPR, HIPAA, PCI-DSS), and digital transformation initiatives increasing the attack surface.

3.2 Spending by Threat Category

The table below summarizes estimated financial impact and defense spending for each of the 25 identified threats, based on published industry research.

Threat	Est. Spending / Impact	Primary Defense Mechanisms	Source
Ransomware	$20B+ global damages (2021)	Offline backups, EDR, patch mgmt, IR retainer	Cybersecurity Ventures 2022
APT	$6B+ enterprise security investment	Threat intel, SIEM, NDR, Zero Trust architecture	Gartner 2023
Supply Chain Attack	$46B projected risk (by 2025)	Vendor risk mgmt, code signing, SCA tools	Gartner/Forrester 2022
Zero-Day Exploits	$12B bug bounty & vuln research market	Patch mgmt, WAF, RASP, threat intelligence feeds	HackerOne 2023
Phishing	$3.8B direct losses; $8.6B defenses	Email filters (SEG), awareness training, MFA	FBI IC3 2022; Proofpoint
Cloud Misconfiguration	$4.1B cloud security market (2023)	CSPM tools, IAM reviews, automated compliance	Gartner 2023
Insider Threat	$15.4M avg annual cost per org	UEBA, PAM, DLP, Zero Trust, employee monitoring	Ponemon Institute 2022
SQL Injection	$6.5B application security market	WAF, SAST/DAST, parameterized queries, pen testing	MarketsandMarkets 2022
DDoS	$2.7B DDoS protection market (2023)	CDN scrubbing, rate limiting, anycast, ISP filtering	Allied Market Research 2023
Credential Stuffing	$1.2B reported losses (2022)	MFA, CAPTCHA, behavioral analytics, cred monitoring	Akamai 2022

Threat	Est. Spending / Impact	Primary Defense Mechanisms	Source
Worms	$54B WannaCry global damage	Network segmentation, patch mgmt, endpoint protection	Cybereason 2021
IoT Exploitation	$6.7B IoT security market (2023)	IoT segmentation, firmware updates, network monitoring	IoT Analytics 2023
API Vulnerabilities	$3.6B API security market (2025 proj.)	API gateways, OWASP API Top 10, runtime protection	Salt Security 2023
Spear Phishing	Included in $8.6B phishing defense	Targeted email security, DMARC/DKIM/SPF, exec protection	Proofpoint 2023
Trojans	Part of $12B endpoint security market	EDR/XDR, application whitelisting, sandboxing	IDC 2022

4. Types of Attacks - Trend Analysis (2015 - 2024)

4.1 Methodology

Attack type percentages represent the approximate proportion of all reported cybersecurity incidents attributed to each attack category per year, derived from synthesizing the following sources: Verizon Data Breach Investigations Report (DBIR) 2015–2024, IBM X-Force Threat Intelligence Index 2020–2024, ENISA Threat Landscape Reports 2018–2023, CrowdStrike Global Threat Report 2021–2024, and Ponemon Institute Cost of a Data Breach Study 2019–2024.

4.2 Year-Wise Trend Table (Selected Attack Types)

Attack Type	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Phishing / Social Eng.	18%	20%	22%	23%	25%	28%	30%	29%	31%	32%
Ransomware	3%	5%	9%	10%	10%	14%	16%	17%	15%	14%
Malware (General)	22%	20%	18%	17%	16%	14%	12%	11%	10%	9%
DDoS Attacks	12%	11%	10%	9%	8%	7%	6%	6%	5%	5%
Web App Attacks	10%	10%	11%	12%	12%	11%	10%	10%	10%	9%
Insider Threats	8%	7%	7%	7%	6%	6%	6%	7%	7%	7%
Supply Chain Attacks	1%	1%	2%	2%	2%	3%	5%	6%	7%	8%
Credential Theft	8%	8%	8%	8%	8%	8%	7%	7%	7%	7%
APT / Espionage	5%	5%	4%	4%	4%	4%	4%	4%	4%	4%
IoT Attacks	1%	2%	3%	4%	4%	4%	4%	4%	4%	4%
BEC	2%	3%	3%	4%	5%	5%	5%	5%	5%	5%
Cloud-Based Attacks	1%	2%	2%	2%	3%	4%	5%	6%	7%	8%
Zero-Day Exploits	4%	3%	3%	3%	3%	3%	3%	3%	3%	3%
Mobile Threats	5%	4%	4%	4%	5%	5%	5%	5%	5%	5%
Cryptojacking	0%	1%	2%	3%	2%	2%	2%	1%	1%	1%

Table 4.1 — Sources: Verizon DBIR 2015–2024, IBM X-Force 2020–2024, ENISA Threat Landscape 2018–2023, CrowdStrike 2021–2024

4.3 Key Trend Observations

Several significant trends emerge from the decade-long analysis:

- Phishing dominance intensifies: Phishing and social engineering attacks grew from 18% of incidents in 2015 to 32% in 2024, making it the single largest and fastest-growing attack category. This reflects the increasing effectiveness of AI-assisted phishing and the persistent challenge of human-layer security.
- Ransomware explosion: Ransomware's share grew from 3% in 2015 to a peak of 17% in 2022 before slightly declining to 14% in 2024. The slight decline reflects improved organizational resilience (backup strategies, incident response) but not a reduction in absolute attack volume.
- Supply chain and cloud attacks surge: Supply chain attacks grew 8-fold (1% to 8%) and cloud-based attacks grew 8-fold (1% to 8%) between 2015 and 2024, reflecting the digital transformation of enterprise infrastructure and the increased targeting of software supply chains.
- General malware declines relatively: General malware incidents declined from 22% to 9% as a proportion of total incidents, not because malware is less prevalent, but because attackers have shifted to more targeted, higher-value techniques (APTs, ransomware, supply chain compromises).
- Cryptojacking peak and stabilization: Cryptojacking emerged sharply in 2017–2018 during the cryptocurrency bull run, peaking at 3% of incidents, then declining as cryptocurrency prices normalized and defenses improved.

5. Data Visualizations

Six visualizations have been produced to illuminate the data collected in this study. Each visualization is described below with its data source and analytical interpretation. All charts are included as embedded visualizations in the accompanying interactive dashboard and as charts within the Excel dataset file.

Visualization 1: Threat Risk Ranking (Bar Chart)

Chart Description & Interpretation

This horizontal bar chart ranks all 25 identified threats by their calculated Risk Score, with color coding by risk level (Critical/High/Medium/Low). The chart clearly shows the gap between the four Critical threats (APT, Zero-Day, Supply Chain, Ransomware — all scoring 56–61) and the remaining threats. Notably, phishing, despite being the most frequent attack type, ranks 5th due to its higher control mitigation rate (55%) compared to APTs (35%). The chart highlights that frequency and risk score are not synonymous — less frequent but harder-to-mitigate threats pose greater organizational risk. Supply Chain Attack achieves the second-highest risk score due to its combination of low control mitigation (30%), high uncertainty (9), and high asset value targeting (10). Source: Author calculations based on NIST SP 800-30 and ENISA Threat Landscape 2023.

Visualization 2: Attack Occurrence Trends (Line Graph)

Chart Description & Interpretation

This multi-line graph plots the percentage of incidents attributed to key attack types from 2015 to 2024. The crossing of trend lines reveals significant shifts in the threat landscape: phishing overtook general malware as the dominant category around 2018–2019. Ransomware's steep rise from 2016 to 2022 is visually dramatic, while its slight plateau in 2023–2024 suggests improved defenses are having marginal effect. Supply chain and cloud attack trajectories show near-identical growth curves (both 1% to 8%), reflecting their shared root cause: trust exploitation in interconnected digital ecosystems. This chart is critical for threat prioritization — security teams should increase investment in the fastest-growing categories. Source: Verizon DBIR 2015–2024, IBM X-Force 2020–2024.

Visualization 3: Distribution of Attack Types (Pie Chart)

Chart Description & Interpretation

This pie chart uses 2024 incident percentage data to show the relative distribution of attack types in the current threat landscape. Phishing/Social Engineering dominates at 32%, followed by Ransomware (14%) and Malware-General (9%). Together, the top three categories account for 55% of all incidents. The chart makes visible the 'long tail' of less frequent but high-impact threats (e.g., Zero-Day Exploits at 3%, APT at 4%) that require disproportionate security investment relative to their incident frequency due to their high potential impact. This distribution reinforces the need for layered security: broad defenses for common threats and specialized controls for high-value, low-frequency attacks. Source: CrowdStrike Global Threat Report 2024, ENISA Threat Landscape 2023.

Visualization 4: Risk Score vs. Security Spending (Scatter Plot)

Chart Description & Interpretation

This scatter plot maps each of the 25 threats on two axes: Risk Score (x-axis) and Global Security Spending in billions USD (y-axis). A positive correlation is expected — higher risk should drive higher spending — but outliers reveal important mismatches. Supply Chain Attacks show high risk (~58) but very high spending ($46B), reflecting the market's recognition of this emerging threat. Ransomware shows high spending ($20B) commensurate with its high risk. Some medium-risk threats (e.g., IoT Exploitation at $6.7B) appear over-invested relative to their risk score, while others (e.g., Vishing at $1B) may be under-resourced. These mismatches suggest opportunities to rebalance security budgets. Source: Author calculations; spending data from Gartner 2023, Cybersecurity Ventures 2022–2024.

Visualization 5: Stacked Bar Chart — Attack Types Across Years

Chart Description & Interpretation

This stacked bar chart shows the compositional change in the attack landscape from 2015 to 2024. Each bar represents 100% of reported incidents in that year, with segments showing each attack type's proportion. The visual shift from 'general malware' dominated stacks (2015–2016) to 'phishing + ransomware + cloud/supply chain' dominated stacks (2022–2024) is striking. The shrinking blue segment (general malware) accompanied by the growing orange/red segments (phishing, ransomware) confirms the weaponization and specialization trend in cybercrime. The chart also shows that while legacy threats shrink proportionally, they do not disappear — they form a persistent baseline. Source: Verizon DBIR 2015–2024, ENISA 2018–2023.

Visualization 6: Defense Mechanism Heatmap

Chart Description & Interpretation

This matrix heatmap maps the 25 identified threats (rows) against 12 primary defense mechanisms (columns), with cell color indicating effectiveness (dark = highly effective, light = minimal effectiveness). Key observations: Multi-Factor Authentication (MFA) is broadly effective across phishing, credential stuffing, and password attacks but provides no protection against zero-day exploits or supply chain attacks. Endpoint Detection & Response (EDR) is highly effective against malware categories but limited against network-based attacks. Network segmentation is uniquely effective at limiting lateral movement in APT and worm scenarios. The heatmap reveals that no single control is universally effective, reinforcing the defense-in-depth principle. It also identifies that supply chain attacks and zero-day exploits have the sparsest defense coverage, validating their Critical risk classification. Source: NIST Cybersecurity Framework, CIS Controls v8, MITRE ATT&CK Mitigations.

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